Ao LIU

Ph.D. in Computer Science

For Applying Principle Applied Scientist at SIEMENS
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Summary

Industry End-to-end experience in designing and deploying xx-billion-parameter models in production. Expertise in bridging theoretical analysis with practical problem-solving in industrial settings. 3 years at Google Core ML & 9 months at MIT-IBM Watson AI Lab

Research 21 publications (9 as first-author, 7 as corresponding-author), 2 patents, and 390+ citations

Expertise Machine Learning (large-small model collaboration, personalization, recommendation systems, LLM, learning to rank), Trustworthy AI (differential privacy, robustness, explainability), Computational Social Choice, and Theoretical Computer Science.

Education

1/2018 – 5/2023 **Ph.D.** in **Computer Science**, Rensselaer Polytechnic Institute (RPI) — Troy, NY, USA
Thesis: *Group Decision Makings from Partial Preferences*

8/2015 – 5/2018 **M.Eng.** in **Material Engineering**, Rensselaer Polytechnic Institute — Troy, NY, USA
Received *Presidential Graduate Research Fellowship*

8/2010 – 5/2014 **B.S.** in **Mathematics and Physics**, Tsinghua University — Beijing, China
Minor in Computer Technology, and in *Academic Talent Program*

Experience (all full-time)

7/2023 – 1/2026 **Machine Learning Engineer** at **Google Core ML** — Mountain View, CA, USA

Project 1: SplitFormer, A Low-Latency, Splittable Transformer Architecture

- Conducted a systematic theoretical analysis demonstrating that prevalent approaches (e.g., FlashAttention) cannot provide meaningful latency reduction for the target model; proposed SplitFormer and proved its theoretical lower bound of 90% inference latency reduction.
- Held end-to-end responsibility for architecture design, implementation, training, and production deployment of SplitFormer, including integration into the *Google Shopping* recommendation system.
- Achieved 3× model parameters and 5% relative accuracy improvement while reducing compute consumption by 30%, corresponding to multi-million-dollar estimated annual revenue contribution on *Google Shopping*.

Project 2: User Behavior Summarization via Tokenization

- Designed and deployed an end-to-end tokenization pipeline that maps user behavior summaries to LLM-compatible token representations, covering model training, token-storage, inference, and low-latency serving.
- Reduced storage footprint by approximately 60% and training cost by 25%, compared to raw embedding baselines.

Project 3: Large Encoder Model for User Behavior Prediction & Summarization

- Designed and productionized its *time embedding* module – from design to system integration – improving prediction accuracy by 17% and summarization accuracy by 4% (relative).
- Contributed to the full model lifecycle, including architecture design, data processing and feature engineering, pre-training and post-training, and online serving.

5/2022 – 8/2022 **Research Intern** at **Google Core ML** — Mountain View, CA, USA

Project: Unbiased Learning to Rank with a More Accurate Position Bias Estimator

- Proposed a novel probabilistic model for estimating and correcting position bias in recommendation systems; designed and trained the corresponding system.
- Outperformed the previous state-of-the-art by approximately 6% in prediction accuracy.

5/2019 – 8/2019 and 9/2018 – 1/2019	**Visiting Scholar** at **MIT-IBM Watson AI Lab** — Cambridge, MA, USA Project: Certifiably Robust Interpretation via Rényi Differential Privacy ○ Proposed the first algorithm with provable robustness against ℓ_∞-norm adversarial attacks. ○ Achieved 12% improvement in robustness over the state-of-the-art, with no additional computational cost and higher accuracy. ○ Results published as a top-tier paper and two U.S. patents.

Skills

Programming	Python, C/C++, TensorFlow, PyTorch, MATLAB
Design	Large-small model collaboration, personalization, LLM, complexity analysis, user modeling, MCMC, privacy analysis, robustness analysis, model interpretability, smoothed analysis

Selected Research Outputs (* denotes corresponding author)

AIJ & AAAI	**Certifiably Robust Interpretation via Rényi Differential Privacy** *Ao Liu**, Xiaoyu Chen, Sijia Liu, Lirong Xia, and Chuang Gan
TMLR	**Smoothed Differential Privacy** *Ao Liu**, Yu-Xiang Wang, and Lirong Xia
AAAI (oral)	**Near-Neighbor Methods in Random Preference Completion** *Ao Liu**, Qiong Wu, Zhenming Liu, and Lirong Xia
AAAI (oral)	**Learning Plackett-Luce Mixture from Partial Preferences** *Ao Liu**, Zhibing Zhao, Chao Liao, Pinyan Lu, and Lirong Xia
AAAI (spotlight)	**The Semi-Random Likelihood of Doctrinal Paradoxes** *Ao Liu** and Lirong Xia
UAI	**Accelerating Voting by Quantum Computation** *Ao Liu**, Qishen Han, Lirong Xia, and Nengkun Yu
ETRA (oral)	**Differential Privacy for Eye-Tracking Data** *Ao Liu**, Lirong Xia, Andrew Duchowski, Reynold Bailey, Kenneth Holmqvist, and Eakta Jain
UAI (oral)	**How Private Are Commonly-Used Voting Rules?** *Ao Liu*, Yun Lu*, Lirong Xia, and Vassilis Zikas
Polymer Physics	**Simulation of Pulse Responses of Lithium Salt-Doped Poly-Ethyleneoxide** *Ao Liu*, F. Zeng*, Y. Hu, S. Lu, W. Dong, X. Li, C. Chang, and D. Guo
US Patent	**Certifiably Robust Interpretation** *Ao Liu*, Sijia Liu, Bo Wu, Lirong Xia, Qi Cheng Li, and Chuang Gan
US Patent	**Interpretation Maps with Guaranteed Robustness** *Ao Liu*, Sijia Liu, Abhishek Bhandwaldar, Chuang Gan, Lirong Xia, and Qi Cheng Li

▶ **See all 21 publications and 2 patents at** [Personal Website] [Google Scholar]

Acdemic Impact

Reviewer	Journal: Information Sciences, ACM ToIS, TMLR, DMLR, IPA, and Sankhya B Conference: NeurIPS, ICML, ICLR, AAAI, IJCAI, UAI, and AISTATS
8/2019	**Invited Talk**, ITCS, Shanghai University of Finance & Economics Topic: *Preference Learning from General Partial Orders*
9/2019 – 5/2022	RPI-IBM **AI Horizon Scholorship** (fund 3 years' tuition and stipend)
9/2016 – 5/2017	**Presidential Graduate Research Fellowship** [Certificate]